Call Log Using Doubly Linked List

Problem Statement

A telephone system keeps a call log in which each entry stores a caller's name and the call duration in minutes.

Two separate call logs must be stored using a **doubly linked list**, where each node has next and prev pointers.

The program must read both logs from the user by appending each call at the end of its list.

The second log must then be merged onto the end of the first, keeping the links correct in both directions.

The merged log must be displayed forward and backward.

The program must search for a caller and report that caller's duration, or report that the caller was not found.

The program must delete a caller's entry and repair the links of the neighbouring nodes, including the head case.

The program must compute the total number of calls and the total call duration.

All dynamically allocated nodes must be freed before the program ends.